

# Finance Chatbot Using Dialogflow and Kommunicate

Project Report — MBA Applied Finance  
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## 1. Objective

The objective of this project is to design, develop, and deploy an AI-powered Finance Chatbot using Google Dialogflow ES and Kommunicate. The chatbot is intended to assist users with common financial queries such as NPV, IRR, EMI calculation, loan advisory, investment guidance, tax saving tips, and budgeting advice — making financial knowledge accessible through a conversational interface.

## 2. Tools Used

| Tool                  | Purpose                                                         |
|-----------------------|-----------------------------------------------------------------|
| Google Dialogflow ES  | Natural Language Understanding (NLU) and Intent Management      |
| Google Cloud Platform | Service Account creation and JSON key generation for API access |
| Kommunicate           | Chatbot deployment, live chat interface, and human handoff      |
| GitHub                | Version control and project submission                          |

## 3. Dialogflow Setup

A new Dialogflow ES agent named 'MBA Finance Bot' was created under the Google Cloud project. The following intents were configured with relevant training phrases and text responses:

- NPV\_Intent — Explains Net Present Value with examples
- IRR\_Intent — Explains Internal Rate of Return
- EMI\_Calculator — Provides EMI calculation guidance
- Loan\_Advisor — Offers loan advisory responses
- Investment\_Advisor — Covers investment planning basics
- Tax\_Saving\_Advisor — Guides on tax-saving instruments
- Budgeting advice — Helps with personal budgeting
- Break\_Even\_Intent — Explains break-even analysis
- Credit\_Score\_Advisor — Advises on credit scores
- Current\_Ratio\_Intent — Explains liquidity ratios
- Debt\_Equity\_Intent — Covers leverage ratios
- ROI\_Intent — Explains Return on Investment
- Insurance\_Advisor — Covers insurance basics

## 4. Google Cloud Configuration

To integrate Dialogflow with Kommunicate, a Service Account was created on Google Cloud Platform with the Dialogflow API Client role. A JSON key was generated and downloaded from the Service Account, which was subsequently uploaded to Kommunicate to authenticate the bot integration.

## 5. Kommunicate Deployment

The Dialogflow bot was integrated into Kommunicate by uploading the JSON service account key. A bot named 'mba-finance-bot' was created and configured with human handoff settings. The live chatbot was tested through the Kommunicate dashboard and a public demo link was generated for submission.

Live Demo Link: <https://www.kommunicate.io/livechat-demo?appId=8a2fab07b0289155890e23bdd3bb7a58&botIds;=mba-finance-bot-1op0q&assignee;=mba-finance-bot-1op0q>

## 6. Testing

The chatbot was tested using Dialogflow's built-in 'Try It Now' console and through the live Kommunicate interface. Test queries included 'What is NPV?', 'Explain IRR', 'How to calculate EMI?', and 'Give me investment advice'. The bot responded correctly to all configured intents and the Default Fallback Intent handled unrecognized queries gracefully.

## 7. Results

The project successfully demonstrates the development of a functional AI-powered finance chatbot. Key outcomes include:

- 13 finance-related intents created and trained in Dialogflow ES
- Accurate intent matching for all major financial query categories
- Successful deployment on Kommunicate with a publicly accessible demo link
- Human handoff feature configured for queries beyond bot capability
- GitHub repository created with full documentation and screenshots

## 8. Conclusion

This project demonstrates the practical application of Conversational AI in the domain of personal finance. By combining Google Dialogflow ES for NLU and Kommunicate for deployment, an accessible and interactive finance assistant was built. The chatbot can serve as a foundation for more advanced financial advisory systems integrating real-time data, user authentication, and personalized recommendations.